

STAGE 0 FOUNDATION

Austin Build Feasibility AI

A staged platform for Austin lot search, zoning-aware feasibility, schematic concepts, AI visuals, and professional review packets.

STAGE 4

Project Intake

Project name

Austin ADU feasibility

Property type

ADU

Risk

Medium

Budget

85000

Units

2

Lot sqft

6500

Building sqft

2200

Beds

4

Baths

2

Validated Contract

```
{
  "spec": {
    "projectName": "Austin ADU feasibility",
    "city": "Austin",
    "state": "TX",
    "propertyType": "adu",
    "totalBudgetUsd": 850000,
    "targetLotSqft": 6500,
    "targetBuildingSqft": 2200,
    "bedrooms": 4,
    "bathrooms": 3,
    "units": 2,
    "stylePreferences": [
      "warm modern",
      "natural light"
    ],
    "riskTolerance": "medium"
  },
  "missingFields": [],
  "assumptions": [
    "Structured intake fields"
```

4



5



validated locally."

Validate Spec**Conversational request**

I want an ADU-friendly warm modern home in Austin with 4 beds, 3 baths, 2 units, 2200 sqft house, 6500 lot sqft, and a budget of 850k.

**Normalize With API****Find Prototype Lots****Ranked Prototype Lots**

Results come from a static Kaggle prototype dataset and are not active listings.

Cross-reference parcel/property identity with Travis/TCAD or official records.

Prototype Lot

100/100

4307 Avenue G, Austin, TX 78751**Hyde Park / Central Austin**

\$825,000 · 6,600 lot sqft · 2 unit

Prototype static dataset row. This is not an active listing or live MLS availability.

View Context**Prototype Lot**

86.42/100

5404 Duval St, Austin, TX 78751**North Loop**

\$1,050,000 · 7,400 lot sqft · 2 unit

Prototype static dataset row. This is not an active listing or live MLS availability.

View Context**Lot Context**

4307 Avenue G, Austin, TX 78751

Hyde Park / Central Austin

Coordinates: 30.298, -97.741

[Open Map](#)

Run Feasibility Check

Prototype listing context only; this is not active inventory.

Parcel boundaries, zoning, and permits require official verification.

No official parcel id is attached to this static fallback row yet.

No zoning feature has been joined to this prototype row yet.

No permit history has been joined to this prototype row yet.

Mapbox token is not configured, so aerial image URL is unavailable.

Compliance Feasibility

4307 Avenue G, Austin, TX 78751

8 advisory findings generated; 6 require review or additional data.

Category	Value	Basis	Status	Confidence	Source
Address Lot information	4307 Avenue G, Austin, TX 78751	known	passes	low	Kaggle Address should I
Part of town Lot information	Hyde Park / Central Austin	known	passes	low	Kaggle
Lot size Lot information	6,600 sqft	known	passes	low	Kaggle check t District Target I
Requested units	2	known	passes	high	User in

Lot information					
Recorded units Lot information	2	known	passes	low	Kaggle Unit co entitlem
Zoning district Zoning	SF-3 (estimated)	estimated	warning	low	Austin C Official 3 is a co placeho Feature
Front setback Setbacks	25 ft estimate	estimated	warning	low	Austin I Advisor against compat
Side setback Setbacks	5 ft estimate	estimated	warning	low	Austin I Advisor setback height, i
Rear setback Setbacks	10 ft estimate	estimated	warning	low	Austin I Advisor with sur
Building square footage Building coverage	2,150 sqft	known	passes	low	Kaggle Target k
FAR / building coverage Building coverage	32.6% existing building-to-lot ratio	estimated	warning	low	Derivec and lot official Existing allowed
Impervious cover Building coverage	44.6% rough impervious- cover estimate	estimated	warning	low	Austin I Rough e plus dri and sur
Tree ordinance risk Environmental	Estimate: medium	public data pending	unknown	low	Austin C arborist Public t city tree need a s

Floodplain / WUI overlays Environmental	Public overlay join pending	public data pending	unknown	low	Austin 1 Feature Feature Use par before c
Crime statistics Public safety	Public incident join pending	public data pending	unknown	low	Austin 0 Join rec and dat categor
Permit history Permitting	Public permit join pending	public data pending	unknown	low	Austin 1 Austin p availabl address

Zoning data unavailable

unknown

No zoning value is attached to this parcel yet.

Confidence: low · Professional verification: required

Advisory feasibility only

warning

This check is a planning screen, not City of Austin approval or a substitute for architect, engineer, survey, legal, arborist, or city review.

Confidence: high · Professional verification: required

Static fallback data

warning

This property comes from a static MVP fallback dataset and must not be treated as active inventory.

Confidence: high · Professional verification: required

Lot size target

passes

Static lot size estimate of 6,600 sqft meets the target of 6,500 sqft.

Confidence: low · Professional verification: required

Unit count target

passes

The record's unit count of 2 meets the requested 2 unit target.

Confidence: low · Professional verification: required

Official zoning join required

unknown

No official zoning district has been joined yet, so allowed use, setbacks, height, impervious cover, and compatibility cannot be confirmed.

Confidence: low · Professional verification: required

Tree review unknown

unknown

Tree constraints cannot be evaluated until a tree survey or official city tree data is joined.

Confidence: low · Professional verification: required

Floodplain and WUI overlays unknown

unknown

Floodplain and Wildland-Urban Interface overlay impacts are not joined yet and must be verified before design decisions.

Confidence: low · Professional verification: required

Generate Schematic Options

Schematic Options

4307 Avenue G, Austin, TX 78751

Stage 6E LLM schematic agent outputs are conceptual planning studies only.

When LLM_SCHEMATIC_ENABLED is false or OpenAI is unavailable, the service falls back to the local generator.

Architect, civil engineer, surveyor, attorney, and city review are required before design reliance.

human_comfort

human comfort

2,200 sqft · 2 units

Two-story duplex-style ADU massing sized to 2,200 sf total, with one 2-bedroom/1.5-bath unit on Level 1 and one 2-bedroom/1.5-bath unit on Level 2. This revision fixes prior program shortfalls by fully providing 4 bedrooms and 3 baths total across 2 units, while improving unit separation through distinct exterior entries and stacked wet-wall/service cores. Warm modern character comes from simple volumes, shaded openings, and open living spaces with strong south/east daylight.

Orient main living/dining spaces and primary glazing toward south/east where

feasible, with deeper overhangs on south and limited west-facing glass. Stack kitchens, baths, and laundry near a central plumbing spine to reduce cost and heat gain from dispersed wet rooms. Bedrooms are placed deeper in plan or toward quieter sides for privacy.

Total building area is targeted at 2,200 sf gross across two floors, roughly 1,100 sf per level.

Because zoning is not joined, this is a schematic feasibility concept only and not a claim of entitlement, permit approval, or code compliance.

North is assumed up; actual placement of entries, stair, and glazing should be adjusted once parcel geometry, setbacks, and tree constraints are verified.

Two units are treated as vertically stacked independent dwellings, each with 2 bedrooms and 1.5 baths, satisfying the requested 4 bedrooms, 3 bathrooms, and 2 units.

Unit separation is improved at concept level with dedicated exterior access, vertically stacked wet cores, and limited shared building zones, but geometric/code-rated separation must be verified in later design.

Room dimensions are approximate schematic sizes intended to reconcile to the target area after wall thicknesses, structure, and mechanical distribution are developed.

Tree review, floodplain/WUI overlay review, impervious cover, parking, height, and setback impacts remain unknown until official city and survey data are joined.

Generated by OpenAI structured output and validated through the local schematic contract.

OpenAI revision loop attempt 3 of 3: deterministic quality checker needs another revision.

Returned best available OpenAI plan after 3 revision attempts; unresolved quality warnings remain advisory.

Level 1

1,100 sqft

Footprint: 40.6 ft x 27.1 ft · OpenAI concept scale: 1 SVG plan unit = 0.41 ft horizontally and 0.27 ft vertically; room dimensions are model-proposed and rounded.

Plan quality

fails · 62/100

Area reconciliation: Room areas reconcile to the target floor-plan square footage.

Footprint area: Footprint dimensions reconcile to the stated floor-plan square footage.

Room bounds: All rooms fit inside the normalized floor-plan footprint.

Room overlap: No meaningful room overlaps detected.

Program fit: Plan is missing requested program elements: 2 bedroom(s), 1

bath(s).

Room dimensions: Room dimensions meet MVP minimum size and proportion assumptions.

Circulation: Plan includes a legible public/private room mix and modeled circulation.

Path connectivity: All rooms are reachable through the modeled connection graph.

Openings: Exterior door and multiple window openings are modeled.

Unit separation: Second-unit separation is incomplete and needs geometric/code refinement.

Solar orientation: Openings favor the assumed south/east daylight strategy.

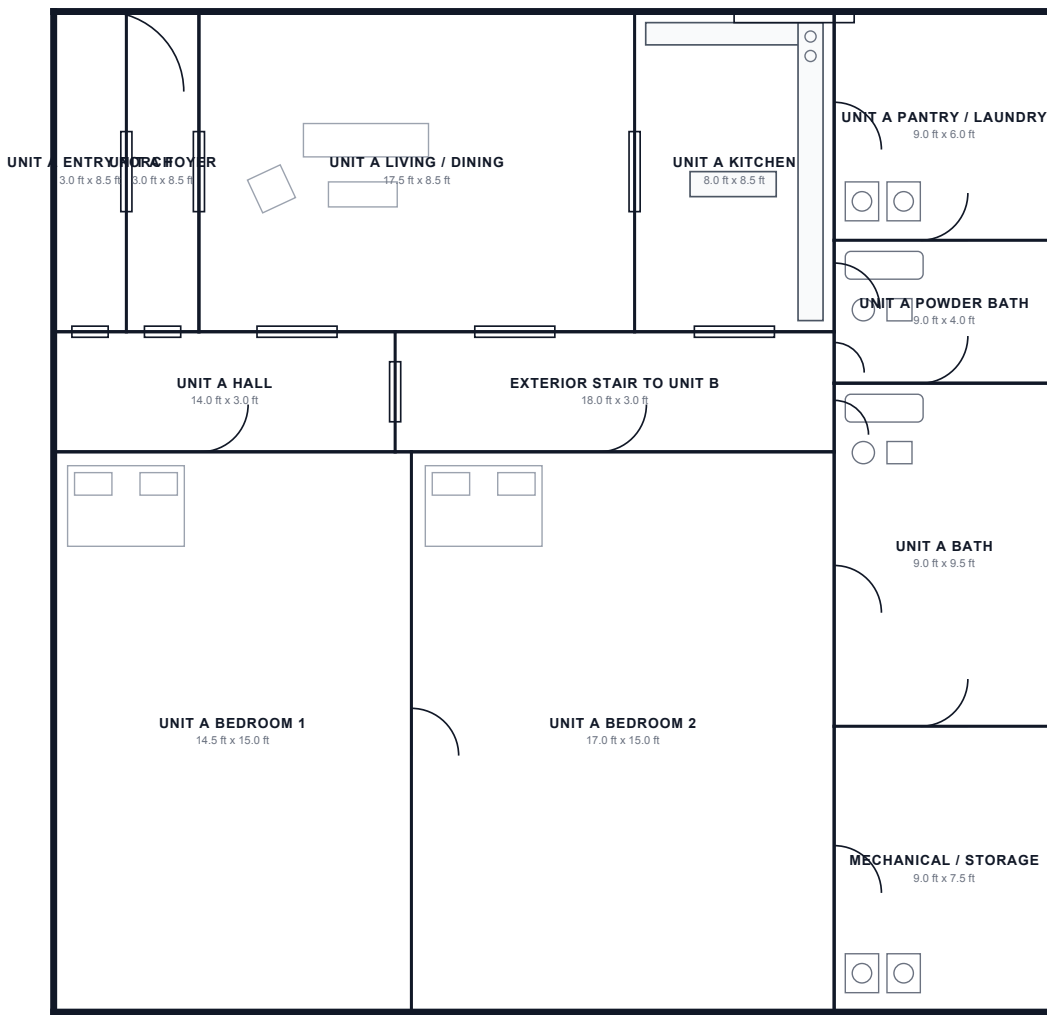
Quality checks are deterministic MVP heuristics for product review, not professional architectural validation.

Stage 6H uses this report with the OpenAI revision loop and structured geometry refiner.

human_comfort - Level 1

1,100 sqft | 40.6 ft x 27.1 ft footprint

OpenAI concept scale: 1 SVG plan unit = 0.41 ft horizontally and 0.27 ft vertically; room dimensions are model-proposed and rounded.



Unit A Entry Porch: 25.54 sqft · 3.0 ft x 8.5 ft

Unit A Foyer: 25.54 sqft · 3.0 ft x 8.5 ft

Unit A Living / Dining: 153.24 sqft · 17.5 ft x 8.5 ft

Unit A Kitchen: 70.24 sqft · 8.0 ft x 8.5 ft

Unit A Pantry / Laundry: 55.31 sqft · 9.0 ft x 6.0 ft

Unit A Powder Bath: 34.58 sqft · 9.0 ft x 4.0 ft

Unit A Hall: 45.05 sqft · 14.0 ft x 3.0 ft

Unit A Bedroom 1: 220.23 sqft · 14.5 ft x 15.0 ft

Unit A Bath: 82.97 sqft · 9.0 ft x 9.5 ft

Unit A Bedroom 2: 260.25 sqft · 17.0 ft x 15.0 ft

Mechanical / Storage: 69.14 sqft · 9.0 ft x 7.5 ft

Exterior Stair to Unit B: 57.91 sqft · 18.0 ft x 3.0 ft

Download OpenAI architectural concept SVG

Orient main living/dining spaces and primary glazing toward south/east where feasible, with deeper overhangs on south and limited west-facing glass. Stack kitchens, baths, and laundry near a central plumbing spine to reduce cost and heat gain from dispersed wet rooms. Bedrooms are placed deeper in plan or toward quieter sides for privacy.

Level 1 is approximately 1,100 sf gross including Unit A and shared/independent access components.

Program fit corrected: this level now includes 2 bedrooms and 1.5 baths for Unit A.

Entry hall minimum-size issue corrected by using a true foyer rather than a narrow hall.

Unit separation improved with distinct exterior access, no required shared interior corridor, and service/core buffering between occupancies.

Level 2

1,100 sqft

Footprint: 40.6 ft x 27.1 ft · OpenAI concept scale: 1 SVG plan unit = 0.41 ft horizontally and 0.27 ft vertically; room dimensions are model-proposed and rounded.

Plan quality

fails · 62/100

Area reconciliation: Room areas reconcile to the target floor-plan square footage

Footage.

Footprint area: Footprint dimensions reconcile to the stated floor-plan square footage.

Room bounds: All rooms fit inside the normalized floor-plan footprint.

Room overlap: No meaningful room overlaps detected.

Program fit: Plan is missing requested program elements: 2 bedroom(s), 1 bath(s).

Room dimensions: Room dimensions meet MVP minimum size and proportion assumptions.

Circulation: Plan includes a legible public/private room mix and modeled circulation.

Path connectivity: All rooms are reachable through the modeled connection graph.

Openings: Exterior door and multiple window openings are modeled.

Unit separation: Second-unit separation is incomplete and needs geometric/code refinement.

Solar orientation: Openings favor the assumed south/east daylight strategy.

Quality checks are deterministic MVP heuristics for product review, not professional architectural validation.

Stage 6H uses this report with the OpenAI revision loop and structured geometry refiner.

human_comfort - Level 2

1,100 sqft | 40.6 ft x 27.1 ft footprint

OpenAI concept scale: 1 SVG plan unit = 0.41 ft horizontally and 0.27 ft vertically; room dimensions are model-proposed and rounded.





Unit B Entry Landing: 28.16 sqft · 3.0 ft x 8.5 ft

Unit B Living / Dining: 168.96 sqft · 19.5 ft x 8.5 ft

Unit B Kitchen: 77.44 sqft · 9.0 ft x 8.5 ft

Unit B Pantry / Laundry: 66.75 sqft · 9.0 ft x 7.5 ft

Unit B Powder Bath: 41.73 sqft · 9.0 ft x 4.5 ft

Unit B Hall: 102.96 sqft · 31.5 ft x 3.0 ft

Unit B Bedroom 1: 149.58 sqft · 19.5 ft x 7.5 ft

Unit B Bath: 100.14 sqft · 9.0 ft x 11.0 ft

Unit B Bedroom 2: 240.24 sqft · 31.5 ft x 7.5 ft

Flex Nook / Study: 90.66 sqft · 12.0 ft x 7.5 ft

Linen / Owner Storage: 33.38 sqft · 9.0 ft x 3.5 ft

Download OpenAI architectural concept SVG

Orient main living/dining spaces and primary glazing toward south/east where feasible, with deeper overhangs on south and limited west-facing glass. Stack kitchens, baths, and laundry near a central plumbing spine to reduce cost and heat gain from dispersed wet rooms. Bedrooms are placed deeper in plan or toward quieter sides for privacy.

Level 2 is approximately 1,100 sf gross for Unit B.

Program fit corrected: this level now includes 2 bedrooms and 1.5 baths for Unit B.

Upper-level social space benefits from better daylight and breezes while limiting west glazing.

Stacked plan reduces construction complexity and supports cleaner warm-modern massing.

6 feasibility flags carried forward.

Workflow



This is advisory feasibility support, not official city approval or professional advice.